Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2012 – Paper 2 – Solutions**

June 2012 – Question 1 – Solution

(a) The expression $f(x) = 2x^3 - px^2 + qx - 10$ is divisible by $x - 1$ and has a remainder -6 when divided by $x + 1$.

Find

(i) the values of the constants p and q [7]

$f(x) = 2x^3 - px^2 + qx - 10$

Since $f(x)$ is divisible by $(x - 1)$, then by the Factor Theorem, $f(1) = 0$.

$f(1) = 2(1)^3 - p(1)^2 + q(1) - 10$

$0 = 2 - p + q - 10$

$0 = -8 - p + q$

$q = p + 8$ → Equation 1

Since $f(x)$ has a remainder -6 when divided by $(x + 1)$, then by the Remainder Theorem, $f(-1) = -6$

$f(-1) = 2(-1)^3 - p(-1)^2 + q(-1) - 10$

$-6 = 2(-1) - p - q - 10$

$-6 = -2 - p - q - 10$

$-6 = -12 - p - q$

$q = -12 + 6 - p$

$q = -6 - p$ → Equation 2

Equating both equations gives:

$p + 8 = -6 - p$

$p + p = -8 - 6$

$2p = -14$

$p = \frac{-14}{2}$

$p = -7$

Substituting $p = -7$ into Equation 1 gives:

$q = -7 + 8$

$= 1$

$\therefore$ The values of the constants are $p = -7$ and $q = 1$.

(ii) the factors of $f(x)$ [3]

$f(x) = 2x^3 - px^2 + qx - 10$

Since $p = -7$ and $q = 1$, then we have,

$f(x) = 2x^3 - (-7)x^2 + (1)x - 10$

$= 2x^3 + 7x^2 + x - 10$

We are given that $f(x)$ is divisible by $(x - 1)$.

By long division,

$$
\begin{array}{r|l}
 & 2x^2 + 9x + 10 \\
\hline
x - 1 & 2x^3 + 7x^2 + x - 10 \\
 & 2x^3 - 2x^2 \\
\hline
 & 9x^2 + x \\
 & 9x^2 - 9x \\
\hline
 & 10x - 10 \\
 & 10x - 10 \\
\hline
 & 0
\end{array}
$$

Now,

$f(x) = (x - 1)(2x^2 + 9x + 10)$

$f(x) = (x - 1)(2x + 5)(x + 2)$

$\therefore$ The factors of $f(x)$ are $(x + 1)$, $(2x + 5)$ and $(x + 2)$.

(b) Find positive integers x and y such that

$$\left(\sqrt{x} + \sqrt{y}\right)^2 = 16 + \sqrt{240}$$ [8]

$$\left(\sqrt{x} + \sqrt{y}\right)^2 = 16 + \sqrt{240}$$

$$\left(\sqrt{x}\right)^2 + 2\left(\sqrt{x}\right)\left(\sqrt{y}\right) + \left(\sqrt{y}\right)^2 = 16 + \sqrt{240}$$

$$x + 2\sqrt{xy} + y = 16 + \sqrt{240}$$

$$x + y + \sqrt{4}\sqrt{xy} = 16 + \sqrt{240}$$

$$x + y + \sqrt{4xy} = 16 + \sqrt{240}$$

Equating terms gives:

$x + y = 16$ → Equation 1

$4xy = 240$ → Equation 2

Rearranging Equation 1 gives:

$y = 16 - x$ → Equation 3

Substituting Equation 3 into Equation 2 gives:

$$4x(16 - x) = 240$$

$$x(16 - x) = 60$$

$$16x - x^2 = 60$$

$$x^2 - 16x + 60 = 0$$

$$(x - 10)(x - 6) = 0$$

Either $x - 10 = 0$ or $x - 6 = 0$

$x = 10$ $x = 6$

When $x = 10$,

$y = 16 - 10$

$= 6$

When $x = 6$,

$y = 16 - 6$

$= 10$

$\therefore x = 10$ and $y = 6$, $x = 6$ and $y = 10$

(c) (i) Solve, for real values of x, the inequality

$|3x - 7| \leq 5$ [5]

$|3x - 7| \leq 5$

Squaring both sides gives:

$$(3x - 7)^2 \leq (5)^2$$

$$9x^2 - 42x + 49 \leq 25$$

$$9x^2 - 42x + 49 - 25 \leq 0$$

$$9x^2 - 42x + 24 \leq 0$$

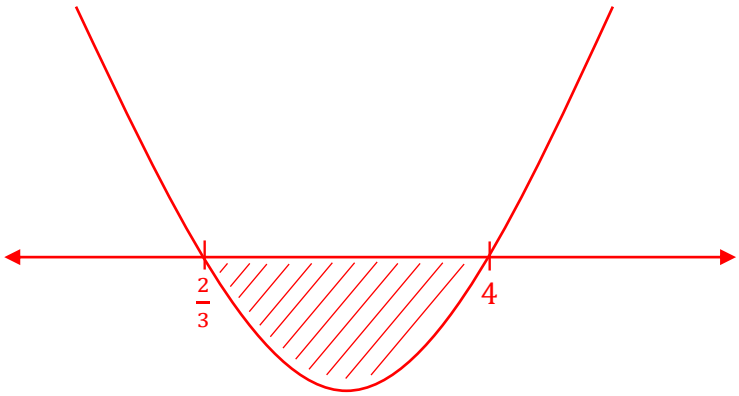
$(\div 3)$

$$3x^2 - 14x + 8 \leq 0$$

$$(3x - 2)(x - 4) \leq 0$$

The critical values are $x = \frac{2}{3}$ and $x = 4$.

The sketch of the graph of $y = 3x^2 - 14x + 8$ is shown below.



Since we are solving $(3x-2)(x-4) \leq 0$, we look where the graph is negative.

The graph is negative for $\frac{2}{3} \leq x \leq 4$.

$\therefore$ The solution set is $\left\{x: \frac{2}{3} \leq x \leq 4\right\}$.

(ii) Show that no real solution, x, exists for the inequality $|3x-7|+5 \leq 0$. [2]

$|3x-7|+5 \leq 0$

$|3x-7| \leq -5$ which is a contradiction.

Note that $|3x-7|$ is always positive or zero.

That is, $|3x-7| \geq 0$.

$\therefore$ In the given equality, no real solution exists for $x \in \mathbb{R}$.

Q.E.D.

Total: 25 marks

June 2012 – Question 2 – Solution

(a) The function f on $\mathbb{R}$ is defined by

$$f: x \to x^2 - 3$$

(i) Find, in terms of x, $f(f(x))$. [3]

$$f(x) = x^2 - 3$$

$$f(f(x)) = f(x^2 - 3)$$

$$= (x^2 - 3)^2 - 3$$

$$= x^4 - 6x^2 + 9 - 3$$

$$= x^4 - 6x^2 + 6$$

$$\therefore f(f(x)) = x^4 - 6x^2 + 6$$

(ii) Determine the values of x for which

$$f(f(x)) = f(x + 3)$$ [6]

$$f(f(x)) = x^4 - 6x^2 + 6$$

Now,

$$f(x + 3) = (x + 3)^2 - 3$$

$$= x^2 + 6x + 9 - 3$$

$$= x^2 + 6x + 6$$

Since $f\big(f(x)\big) = f(x+3)$, then

$$x^4 - 6x^2 + 6 = x^2 + 6x + 6$$

$$x^4 - 6x^2 - x^2 - 6x + 6 - 6 = 0$$

$$x^4 - 7x^2 - 6x = 0$$

$$x(x^3 - 7x - 6) = 0$$

Let $g(x) = x^3 - 7x - 6$.

Suppose $x = -1$. Then,

$$g(-1) = (-1)^3 - 7(-1) - 6$$

$$= -1 + 7 - 6$$

$$= 0$$

Since $g(-1) = 0$, then by the Factor Theorem, $(x+1)$ is a factor of $g(x)$.

By long division,

$$\begin{array}{r|l}
 & x^2 - x - 6 \\
\hline
x + 1 & x^3 + 0x^2 - 7x - 6 \\
 & x^3 + x^2 \\
\cline{2-2}
 & -x^2 - 7x \\
 & -x^2 - x \\
\cline{2-2}
 & -6x - 6 \\
 & -6x - 6 \\
\cline{2-2}
 & 0 \\
\cline{2-2}
\end{array}$$

Now,

$g(x) = x^3 - 7x - 6$

$= (x+1)(x^2 - x - 6)$

$= (x+1)(x-3)(x+2)$

Hence,

$x(x^3 - 7x - 6) = 0$

$x(x+1)(x-3)(x+2) = 0$

Either $x = 0$, $x + 1 = 0$, $x - 3 = 0$ or $x + 2 = 0$

$x = -1$ $x = 3$ $x = -2$

$\therefore$ The values of x are $x = -2, -1, 0, 3$.

(b) The roots of the equation $4x^2 - 3x + 1 = 0$ are α and β.

Without solving the equation

(i) write down the values of $\alpha + \beta$ and $\alpha\beta$ [2]

$4x^2 - 3x + 1 = 0$ which is in the form $ax^2 + bx + c = 0$,

where $a = 4, b = -3$ and $c = 1$.

$\alpha + \beta = -\frac{b}{a}$

$= \frac{-(-3)}{4}$

$= \frac{3}{4}$

$\alpha\beta = \frac{c}{a}$

$= \frac{1}{4}$

$\therefore \alpha + \beta = \frac{3}{4}$ and $\alpha\beta = \frac{1}{4}$

(ii) find the value of $\alpha^2 + \beta^2$ [2]

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(\frac{3}{4}\right)^2 - 2\left(\frac{1}{4}\right)$

$= \frac{9}{16} - \frac{2}{4}$

$= \frac{9}{16} - \frac{8}{16}$

$= \frac{1}{16}$

(iii) obtain a quadratic equation whose roots are $\frac{2}{\alpha^2}$ and $\frac{2}{\beta^2}$ [5]

For the new roots:

Sum of roots $= \frac{2}{\alpha^2} + \frac{2}{\beta^2}$

$= \frac{2\beta^2 + 2\alpha^2}{\alpha^2\beta^2}$

$= \frac{2(\alpha^2 + \beta^2)}{(\alpha\beta)^2}$

$= \frac{2\left(\frac{1}{16}\right)}{\left(\frac{1}{4}\right)^2}$

$= \frac{2\left(\frac{1}{16}\right)}{\left(\frac{1}{16}\right)}$

$= 2$

Product of roots $= \left(\frac{2}{\alpha^2}\right)\left(\frac{2}{\beta^2}\right)$

$$= \frac{4}{\alpha^2\beta^2}$$

$$= \frac{4}{(\alpha\beta)^2}$$

$$= \frac{4}{\left(\frac{1}{4}\right)^2}$$

$$= \frac{4}{\left(\frac{1}{16}\right)}$$

$$= 4 \div \frac{1}{16}$$

$$= 4 \times 16$$

$$= 64$$

Now,

$$x^2 - (sum\ of\ roots)x + (product\ of\ roots) = 0$$

$$x^2 - \left(\frac{2}{\alpha^2} + \frac{2}{\beta^2}\right)x + \left(\frac{2}{\alpha^2} \times \frac{2}{\beta^2}\right) = 0$$

$$x^2 - 2x + 64 = 0$$

$\therefore$ The new equation whose roots are $\frac{2}{\alpha^2}$ and $\frac{2}{\beta^2}$ is:

$$x^2 - 2x + 64 = 0$$

(c) Without the use of calculators or tables, evaluate

(i) $\log_{10}\left(\frac{1}{3}\right) + \log_{10}\left(\frac{3}{5}\right) + \log_{10}\left(\frac{5}{7}\right) + \log_{10}\left(\frac{7}{9}\right) + \log_{10}\left(\frac{9}{10}\right)$ [3]

$$\log_{10}\left(\frac{1}{3}\right) + \log_{10}\left(\frac{3}{5}\right) + \log_{10}\left(\frac{5}{7}\right) + \log_{10}\left(\frac{7}{9}\right) + \log_{10}\left(\frac{9}{10}\right)$$

$$= \log_{10}\left(\frac{1}{3} \times \frac{3}{5} \times \frac{5}{7} \times \frac{7}{9} \times \frac{9}{10}\right)$$

$$= \log_{10}\left(\frac{1}{10}\right)$$

$$= \log_{10}(10^{-1})$$

$$= (-1)\log_{10}(10)$$

$$= (-1)(1)$$

$$= -1$$

(ii) $\sum_{r=1}^{99} \log_{10}\left(\frac{r}{r+1}\right)$ [4]

$$\sum_{r=1}^{99} \log_{10}\left(\frac{r}{r+1}\right)$$

$$= \log_{10}\left(\frac{1}{2}\right) + \log_{10}\left(\frac{2}{3}\right) + \log_{10}\left(\frac{3}{4}\right) + \cdots + \log_{10}\left(\frac{99}{100}\right)$$

$$= \log_{10}\left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \ldots \times \frac{99}{100}\right)$$

$$= \log_{10}\left(\frac{1}{100}\right)$$

$$= \log_{10}(10^{-2})$$

$$= (-2)\log_{10}(10)$$

$$= (-2)(1)$$

$$= -2$$

Total: 25 marks

June 2012 – Question 3 – Solution

(a) (i) Given that $\cos(A+B) = \cos A \cos B - \sin A \sin B$ and $\cos 2\theta = 2\cos^2\theta - 1$,

prove that

$$\cos 3\theta \equiv 2\cos\theta\left[\cos^2\theta - \sin^2\theta - \frac{1}{2}\right] \qquad [7]$$

Required to prove that $\cos 3\theta \equiv 2\cos\theta\left[\cos^2\theta - \sin^2\theta - \frac{1}{2}\right]$.

Taking L.H.S:

$$\cos 3\theta = \cos(2\theta + \theta)$$

$$= \cos 2\theta \cos\theta - \sin 2\theta \sin\theta$$

$$= (2\cos^2\theta - 1)\cos\theta - (2\sin\theta\cos\theta)\sin\theta$$

$$= 2\cos^3\theta - \cos\theta - 2\sin^2\theta\cos\theta$$

$$= 2\cos\theta\left[\cos^2\theta - \frac{1}{2} - \sin^2\theta\right]$$

$$= 2\cos\theta\left[\cos^2\theta - \sin^2\theta - \frac{1}{2}\right]$$

$$= \text{R.H.S.}$$

$$\therefore \cos 3\theta \equiv 2\cos\theta\left[\cos^2\theta - \sin^2\theta - \frac{1}{2}\right]$$

Q.E.D.

(ii) Using the appropriate formula, show that

$$\frac{1}{2}[\sin 6\theta - \sin 2\theta] \equiv (2\cos^2 2\theta - 1)\sin 2\theta \qquad [5]$$

Recall:

$\sin P - \sin Q = 2\cos\left(\frac{P+Q}{2}\right)\sin\left(\frac{P-Q}{2}\right)$

Required to show that $\frac{1}{2}[\sin 6\theta - \sin 2\theta] \equiv (2\cos^2 2\theta - 1)\sin 2\theta$.

Taking L.H.S:

$$\frac{1}{2}[\sin 6\theta - \sin 2\theta] = \frac{1}{2}\left[2\cos\left(\frac{6\theta+2\theta}{2}\right)\sin\left(\frac{6\theta-2\theta}{2}\right)\right]$$

$$= \cos\left(\frac{8\theta}{2}\right)\sin\left(\frac{4\theta}{2}\right)$$

$$= \cos 4\theta \sin 2\theta$$

$$= \cos 2(2\theta)\sin 2\theta$$

$$= (2\cos^2 2\theta - 1)\sin 2\theta$$

$$= \text{R.H.S.}$$

$\therefore \frac{1}{2}[\sin 6\theta - \sin 2\theta] \equiv (2\cos^2 2\theta - 1)\sin 2\theta$

Q.E.D.

(iii) Hence, or otherwise, solve $\sin 6\theta - \sin 2\theta = 0$ for $0 \le \theta \le \frac{\pi}{2}$. [5]

$$\sin 6\theta - \sin 2\theta = 0$$

$$2\left(\frac{1}{2}\right)(\sin 6\theta - \sin 2\theta) = 0$$

$$2(2\cos^2 2\theta - 1)\sin 2\theta = 0$$

$$(2\cos^2 2\theta - 1)(\sin 2\theta) = 0$$

Either $2\cos^2 2\theta - 1 = 0$ or $\sin 2\theta = 0$

$$2\cos^2 2\theta = 1$$

$$\cos^2 2\theta = \frac{1}{2}$$

$$\cos 2\theta = \pm\frac{1}{\sqrt{2}}$$

When $\cos 2\theta = \frac{1}{\sqrt{2}}$,

$$2\theta = \frac{\pi}{4}$$

$$\theta = \frac{\pi}{8}$$

When $\cos 2\theta = -\frac{1}{\sqrt{2}}$,

$$2\theta = \frac{3\pi}{4}$$

$$\theta = \frac{3\pi}{8}$$

When $\sin 2\theta = 0$,

$$2\theta = 0, \pi$$

$$\theta = 0, \frac{\pi}{2}$$

$\therefore \theta = 0, \frac{\pi}{8}, \frac{3\pi}{8}, \frac{\pi}{2}$ for $0 \le \theta \le \frac{\pi}{2}$

(b) Find ALL possible values of $\cos\theta$ such that $2\cot^2\theta + \cos\theta = 0$. [8]

$$2\cot^2\theta + \cos\theta = 0$$

$$2\left(\frac{\cos^2\theta}{\sin^2\theta}\right) + \cos\theta = 0$$

$$\frac{2\cos^2\theta}{\sin^2\theta} + \frac{\cos\theta}{1} = 0$$

$$\frac{2\cos^2\theta + \cos\theta\sin^2\theta}{\sin^2\theta} = 0$$

$$2\cos^2\theta + \cos\theta\sin^2\theta = 0$$

$$2\cos^2\theta + \cos\theta\,(1 - \cos^2\theta) = 0$$

$$2\cos^2\theta + \cos\theta - \cos^3\theta = 0$$

$$\cos^3\theta - 2\cos^2\theta - \cos\theta = 0$$

$$\cos\theta\,(\cos^2\theta - 2\cos\theta - 1) = 0$$

Either $\cos\theta = 0$ or $\cos^2\theta - 2\cos\theta - 1 = 0$

Consider $\cos^2\theta - 2\cos\theta - 1 = 0$,

which is in the form $ax^2 + bx + c = 0$,

where $a = 1, b = -2$ and $c = -1$.

Using the quadratic formula,

$$\cos\theta = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{-(-2) \pm \sqrt{(-2)^2-4(1)(-1)}}{2(1)}$$

$$= \frac{2 \pm \sqrt{4+4}}{2}$$

$$= \frac{2 \pm \sqrt{8}}{2}$$

$$= \frac{2 \pm 2\sqrt{2}}{2}$$

$$= \frac{2(1 \pm \sqrt{2})}{2}$$

$$= 1 \pm \sqrt{2}$$

$\therefore$ All possible values of $\cos\theta$ are $\cos\theta = 0, \cos\theta = 1 - \sqrt{2}$ for $-1 \leq \cos\theta \leq 1$.

Total: 25 marks

June 2012 – Question 4 – Solution

(a)(i) Determine the Cartesian equation of the curve, C, defined by the parametric equations $y = 3\sec\theta$ and $x = 3\tan\theta$. [5]

$y = 3\sec\theta$ and $x = 3\tan\theta$

$\frac{y}{3} = \sec\theta$ $\frac{x}{3} = \tan\theta$

Now,

$\tan^2\theta + 1 = \sec^2\theta$

$\left(\frac{x}{3}\right)^2 + 1 = \left(\frac{y}{3}\right)^2$

$\frac{x^2}{9} + 1 = \frac{y^2}{9}$

$(\times 9)$

$x^2 + 9 = y^2$

$\therefore$ The Cartesian equation of the curve, C, is $x^2 + 9 = y^2$.

(ii) Find the points of intersection of the curve $y = \sqrt{10x}$ with C. [9]

$x^2 + 9 = y^2$ $\rightarrow$ Equation 1

$y = \sqrt{10x}$ $\rightarrow$ Equation 2

Squaring Equation 2 gives:

$y^2 = 10x$ $\rightarrow$ Equation 3

Substituting Equation 3 into Equation 1 gives:

$$x^2 + 9 = 10x$$

$$x^2 - 10x + 9 = 0$$

$$(x - 1)(x - 9) = 0$$

Either $x - 1 = 0$ or $x - 9 = 0$

$x = 1$ $x = 9$

When $x = 1$,

$$y = \sqrt{10(1)}$$

$$= \sqrt{10}$$

When $x = 9$,

$$y = \sqrt{10(9)}$$

$$= \sqrt{90}$$

$$= \sqrt{9 \times 10}$$

$$= \sqrt{9}\sqrt{10}$$

$$= 3\sqrt{10}$$

$\therefore$ The points of intersection are $(1, \sqrt{10})$ and $(9, 3\sqrt{10})$.

(b) Let $\boldsymbol{p}$ and $\boldsymbol{q}$ be two position vectors with endpoints $(-3, 4)$ and $(-1, 6)$ respectively.

(i) Express $\boldsymbol{p}$ and $\boldsymbol{q}$ in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}}$. [2]

$\boldsymbol{p} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$

$\boldsymbol{p} = -3\hat{\boldsymbol{\imath}} + 4\hat{\boldsymbol{\jmath}}$ which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}}$,

where $x = -3$ and $y = 4$.

$\boldsymbol{q} = \begin{pmatrix} -1 \\ 6 \end{pmatrix}$

$\boldsymbol{q} = -\hat{\boldsymbol{\imath}} + 6\hat{\boldsymbol{\jmath}}$ which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}}$,

where $x = -1$ and $y = 6$.

(ii) Obtain the vector $\boldsymbol{p} - \boldsymbol{q}$. [2]

$$\boldsymbol{p} - \boldsymbol{q} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} - \begin{pmatrix} -1 \\ 6 \end{pmatrix}$$

$$= \begin{pmatrix} -3 - (-1) \\ 4 - 6 \end{pmatrix}$$

$$= \begin{pmatrix} -3 + 1 \\ 4 - 6 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ -2 \end{pmatrix}$$

$= -2\hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}}$ which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}}$,

where $x = -2$ and $y = -2$.

(iii) Calculate $\boldsymbol{p} \cdot \boldsymbol{q}$. [2]

$$\boldsymbol{p} \cdot \boldsymbol{q} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 6 \end{pmatrix}$$

$$= (-3)(-1) + (4)(6)$$

$$= 3 + 24$$

$$= 27$$

(iv) Let the angle between $\boldsymbol{p}$ and $\boldsymbol{q}$ be θ. Use the result of (iii) above to calculate θ in degrees. [5]

Given $\boldsymbol{p} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$,

$$|\boldsymbol{p}| = \sqrt{(-3)^2 + (4)^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$= 5$$

Given $\boldsymbol{q} = \begin{pmatrix} -1 \\ 6 \end{pmatrix}$,

$$|\boldsymbol{q}| = \sqrt{(-1)^2 + (6)^2}$$

$$= \sqrt{1 + 36}$$

$$= \sqrt{37}$$

Also, $\boldsymbol{p} \cdot \boldsymbol{q} = 27$.

Now,

$$\cos\theta = \frac{\boldsymbol{p}\cdot\boldsymbol{q}}{|\boldsymbol{p}||\boldsymbol{q}|}$$

$$\cos\theta = \frac{27}{5\sqrt{37}}$$

$$\theta = \cos^{-1}\left(\frac{27}{5\sqrt{37}}\right)$$

$$\theta = 27.4° \qquad \text{(to 1 decimal place)}$$

Total: 25 marks

June 2012 – Question 5 – Solution

(a) (i) Find the values of x for which $\frac{x^3+8}{x^2-4}$ is discontinuous. [2]

Let $f(x) = \frac{x^3+8}{x^2-4}$.

$f(x)$ is discontinuous when the denominator is equal to zero.

So, we have,

$$x^2 - 4 = 0$$

$$(x+2)(x-2) = 0$$

Either $x + 2 = 0$ or $x - 2 = 0$

$x = -2$ $x = 2$

$\therefore$ The values of x for which $\frac{x^3+8}{x^2-4}$ is discontinuous is $x = -2$ and $x = 2$.

(ii) Hence, or otherwise, find

$$\lim_{x \to -2} \frac{x^3+8}{x^2-4}$$ [3]

$$\lim_{x \to -2} \frac{x^3+8}{x^2-4} = \lim_{x \to -2} \frac{(x)^3-(-2)^3}{x^2-4}$$

$$= \lim_{x \to -2} \frac{(x+2)(x^2-2x+4)}{(x+2)(x-2)}$$

$$= \lim_{x \to -2} \frac{(x^2-2x+4)}{(x-2)}$$

$$= \frac{(-2)^2-2(-2)+4}{(-2)-2}$$

$$= \frac{4+4+4}{-4}$$

$$= \frac{12}{-4}$$

$$= -3$$

$$\therefore \lim_{x \to -2} \frac{x^3+8}{x^2-4} = -3$$

(iii) By using the fact that $\lim_{x\to 0} \frac{\sin x}{x} = 1$, or otherwise, find,

$$\lim_{x\to 0} \frac{2x^3+4x}{\sin 2x}$$ [5]

$$\lim_{x\to 0} \frac{2x^3+4x}{\sin 2x} = \lim_{x\to 0} \frac{2x(x^2+2)}{\sin 2x}$$

$$= \lim_{x\to 0} (x^2+2) \cdot \frac{2x}{\sin 2x}$$

$$= \lim_{x\to 0} (x^2+2) \div \frac{\sin 2x}{2x}$$

$$= \lim_{x\to 0} (x^2+2) \div \lim_{x\to 0} \frac{\sin 2x}{2x}$$

$$= \lim_{x\to 0} (x^2+2) \div (1)$$

$$= \lim_{x\to 0} (x^2+2)$$

$$= (0)^2 + 2$$

$$= 0 + 2$$

$$= 2$$

$$\therefore \lim_{x\to 0} \frac{2x^3+4x}{\sin 2x} = 2$$

(b) The function f on $\mathbb{R}$ is defined by

$$f(x) = \begin{cases} x^2 + 1, & x > 1 \\ 4 + px, & x < 1 \end{cases}$$

(i) Find

(a) $\lim_{x \to 1^+} f(x)$ [2]

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (x^2 + 1)$$

$$= (1)^2 + 1$$

$$= 1 + 1$$

$$= 2$$

(b) the value of the constant p such that $\lim_{x \to 1} f(x)$ exists. [4]

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (4 + px)$$

$$= 4 + p(1)$$

$$= 4 + p$$

Since $\lim_{x \to 1} f(x)$ exists, then $\lim_{x \to 1^+} f(x) = \lim_{x \to 1^-} f(x)$.

Hence,

$$2 = 4 + p$$

$$2 - 4 = p$$

$$-2 = p$$

$\therefore$ The value of the constant $p = -2$.

(ii) Hence, determine the value of $f(1)$ for f to be continuous at the point $x = 1$. [1]

Since $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x) = 2$, then $\lim_{x \to 1} f(x) = 2$.

In order for f to be continuous at the point $x = 1$, then $\lim_{x \to 1} f(x) = f(1)$.

Hence, $f(1) = 2$.

(c) A chemical process in a manufacturing plant is controlled by the function

$$M = ut^2 + \frac{v}{t^2}$$

where u and v are constants.

Given that $M = -1$ when $t = 1$ and that the rate of change of M with respect to t is $\frac{35}{4}$ when $t = 2$, find the values of u and v. [8]

$M = ut^2 + \frac{v}{t^2}$

When $t = 1$, $M = -1$.

So, we have,

$(-1) = u(1)^2 + \frac{v}{(1)^2}$

$-1 = u + v$

Now,

$M = ut^2 + \frac{v}{t^2}$

$M = ut^2 + vt^{-2}$

$\frac{dM}{dt} = 2ut - 2vt^{-3}$

When $t = 2, \frac{dM}{dt} = \frac{35}{4}$.

So, we have

$\frac{35}{4} = 2u(2) - 2v(2)^{-3}$

$\frac{35}{4} = 4u - 2v\left(\frac{1}{8}\right)$

$\frac{35}{4} = 4u - \frac{v}{4}$

$35 = 16u - v$

We have obtained two equations:

$-1 = u + v$ → Equation 1

$35 = 16u - v$ → Equation 2

Adding Equation 1 and Equation 2 gives:

$34 = 17u$

$u = \frac{34}{17}$

$u = 2$

Substituting $u = 2$ into Equation 1 gives:

$-1 = 2 + v$

$v = -1 - 2$

$v = -3$

$\therefore$ The values of u and v are $u = 2$ and $v = -3$.

Total: 25 marks

June 2012 – Question 6 – Solution

(a) (i) Given that $y = \sqrt{4x^2 - 7}$, show that $y\frac{dy}{dx} = 4x$. [3]

$$y = \sqrt{4x^2 - 7}$$

$$y = (4x^2 - 7)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(8x)(4x^2 - 7)^{\frac{1}{2}-1}$$

$$\frac{dy}{dx} = 4x(4x^2 - 7)^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{4x}{(4x^2-7)^{\frac{1}{2}}}$$

$$\frac{dy}{dx} = \frac{4x}{\sqrt{4x^2-7}}$$

Required to show that $y\frac{dy}{dx} = 4x$.

Taking L.H.S:

$$y\frac{dy}{dx} = \sqrt{4x^2 - 7} \cdot \frac{4x}{\sqrt{4x^2-7}}$$

$$= 4x$$

$$= \text{R.H.S.}$$

$$\therefore y\frac{dy}{dx} = 4x$$

Q.E.D.

(ii) Hence, or otherwise, show that

$$y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=4 \qquad [3]$$

$\frac{dy}{dx}=4x(4x^2-7)^{-\frac{1}{2}}$

Let $u=4x$, $v=(4x^2-7)^{-\frac{1}{2}}$

$\frac{du}{dx}=4$ $\frac{dv}{dx}=\left(-\frac{1}{2}\right)(8x)(4x^2-7)^{-\frac{1}{2}-1}$

$\frac{dv}{dx}=-4x(4x^2-7)^{-\frac{3}{2}}$

Using the product rule,

$\frac{d^2y}{dx^2}=u\frac{dv}{dx}+v\frac{du}{dx}$

$=(4x)(-4x)(4x^2-7)^{-\frac{3}{2}}+(4x^2-7)^{-\frac{1}{2}}(4)$

$=-16x^2(4x^2-7)^{-\frac{3}{2}}+4(4x^2-7)^{-\frac{1}{2}}$

Required to show that $y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=4$.

Taking L.H.S:

$y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=(4x^2-7)^{\frac{1}{2}}\left[-16x^2(4x^2-7)^{-\frac{3}{2}}+4(4x^2-7)^{-\frac{1}{2}}\right]+\left[4x(4x^2-7)^{-\frac{1}{2}}\right]^2$

$=-16x^2(4x^2-7)^{-1}+4+16x^2(4x^2-7)^{-1}$

$=4$

$=$ R.H.S.

$\therefore y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=4$

Q.E.D.

(b) The curve, C, passes through the point $(-1, 0)$ and its gradient at the point (x, y) is given by

$$\frac{dy}{dx} = 3x^2 - 6x$$

(i) Find the equation of C. [4]

$$\frac{dy}{dx} = 3x^2 - 6x$$

$$y = \int \frac{dy}{dx}\, dx$$

$$y = \int (3x^2 - 6x)\, dx$$

$$y = \frac{3x^3}{3} - \frac{6x^2}{2} + c$$

$$y = x^3 - 3x^2 + c \qquad \rightarrow \text{Equation 1}$$

Substituting the point $(-1, 0)$ into Equation 1 gives:

$$(0) = (-1)^3 - 3(-1)^2 + c$$

$$0 = -1 - 3 + c$$

$$0 = -4 + c$$

$$4 = c$$

$\therefore$ The equation of C is $y = x^3 - 3x^2 + 4$.

(ii) Find the coordinates of the stationary points of C. [3]

At stationary points, $\frac{dy}{dx} = 0$.

When $\frac{dy}{dx} = 0$,

$3x^2 - 6x = 0$

$3x(x-2) = 0$

Either $3x = 0$ or $x - 2 = 0$

$x = 0$ $x = 2$

When $x = 0$,

$y = (0)^3 - 3(0)^2 + 4$

$= 0 - 0 + 4$

$= 4$

When $x = 2$,

$y = (2)^3 - 3(2)^2 + 4$

$= 8 - 3(4) + 4$

$= 8 - 12 + 4$

$= 0$

$\therefore$ The coordinates of the stationary points are $(0, 4)$ and $(2, 0)$.

(iii) Determine the nature of EACH stationary point. [3]

$\frac{dy}{dx} = 3x^2 - 6x$

$\frac{d^2y}{dx^2} = 6x - 6$

When $x = 0$,

$\frac{d^2y}{dx^2} = 6(0) - 6$

$= 0 - 6$

$= -6 \quad (< 0)$

Since $\frac{d^2y}{dx^2} < 0$, then $(0, 4)$ is a maximum point.

When $x = 2$,

$\frac{d^2y}{dx^2} = 6(2) - 6$

$= 12 - 6$

$= 6 \quad (> 0)$

Since $\frac{d^2y}{dx^2} > 0$, then $(2, 0)$ is a minimum point.

(iv) Find the coordinates of the points P and Q at which the curve C meets the x-axis. [5]

Let $f(x) = x^3 - 3x^2 + 4$.

By trial and error, consider $x = -1$.

$f(-1) = (-1)^3 - 3(-1)^2 + 4$

$= -1 - 3 + 4$

$= 0$

Since $f(-1) = 0$, then by the Factor Theorem, $(x+1)$ is a factor of $f(x)$.

By long division,

$$
\begin{array}{r|l}
 & x^2 - 4x + 4 \\
\hline
x + 1 & x^3 - 3x^2 + 0x + 4 \\
 & x^3 + x^2 \\
\hline
 & -4x^2 + 0x \\
 & -4x^2 - 4x \\
\hline
 & 4x + 4 \\
 & 4x + 4 \\
\hline
 & 0 \\
\hline
\end{array}
$$

So, we have,

$f(x) = (x+1)(x^2 - 4x + 4)$

$f(x) = (x+1)(x-2)^2$

The curve cuts the x-axis at $y = 0$.

When $f(x) = 0$,

$(x+1)(x-2)^2 = 0$

Either $\quad x + 1 = 0 \quad$ or $\quad (x-2)^2 = 0$

$x = -1 \qquad x - 2 = 0$

$x = 2$

$\therefore$ The curve meets the x-axis at $P(-1, 0)$ and $Q(2, 0)$.

(v) Hence, sketch the curve C, showing

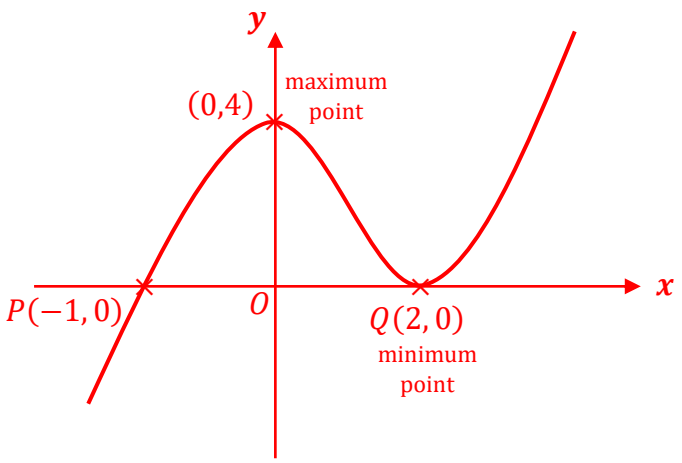
(a) the stationary points

(b) the points P and Q [4]

The x-intercepts are $P(-1, 0)$ and $Q(2, 0)$.

The y-intercept is $(0, 4)$.

The point $(0, 4)$ is a maximum point and the point $(2, 0)$ is a minimum point.



Total: 25 marks